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Accurate Time Series Forecasting of the Saudi Arabian Stock Market Using Hybrid Deep Learning Models

Qassim University | College of Computer | Dept. of Computer Science

Master's Thesis — Based

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1 Introduction

Background

Financial markets are both a mirror and a motor of economic activity — they aggregate dispersed information and transform investor expectations into price movements that shape macroeconomic stability. The Saudi Arabian stock market (TASI) is the largest exchange in the MENA region and a cornerstone of Saudi Vision 2030.

Why TASI?

TASI's unique characteristics make it a compelling forecasting challenge: heavy dependence on global oil prices, periodic structural reforms (foreign investor access 2015, MSCI/FTSE inclusion 2019), and persistent volatility clustering during crises such as the 2014–15 oil collapse and COVID-19.

Research Gap

- Most prior TASI studies use short data windows predating 2020 reforms
- Few apply hybrid deep learning or rolling-window validation
- No study covers the full 2014–2024 horizon including COVID-19 and oil-price crises
- Single-metric evaluation (RMSE only) misses the full error picture

2 Research Objectives

Overarching Aim

Develop and evaluate a hybrid CNN–LSTM deep learning framework for accurate forecasting of TASI using ten years of daily data (2014–2024).

Specific Objectives

① Collect and preprocess daily TASI data (2014–2024): log-returns, ADF stationarity test, PACF analysis, volatility clustering via rolling variance
② Establish econometric benchmarks — ARIMA(1,1,1) and GARCH(1,1) — as interpretable baselines
③ Design and implement standalone LSTM and CNN architectures to assess individual capabilities
④ Construct a hybrid CNN–LSTM architecture optimised through cross-validation
⑤ Evaluate all models using MAE, RMSE, and MAPE with rolling-window validation and crisis-period sub-samples
⑥ Interpret results and derive practical implications for investors, regulators, and policymakers under Vision 2030

3 Methodology

Dataset

Daily TASI closing prices, January 2014 – December 2024 (~2,600 observations). Log-returns computed as consecutive log-differences. Missing values (trading holidays) forward-filled. Outliers retained — they represent genuine market shocks (e.g. COVID-19 crash, oil collapse).

Descriptive Statistics

Mean: 0.000123 · Std Dev: 0.010604 · Min: −8.32% · Max: +8.92%

Skewness: −0.75 (negative shocks dominant) · Kurtosis: 10.69 (heavy tails)

Preprocessing

ADF test — unit root rejected at 1% significance · PACF shows significant autocorrelation at lags 1–2 · 30/60/90-day rolling variance confirms persistent volatility clustering

Five Models Evaluated

① ARIMA(1,1,1) — linear baseline
② GARCH(1,1) — volatility baseline
③ LSTM — 64 units, 60-day lookback, dropout 0.2, Adam optimizer, batch 32
④ CNN — 64 filters, kernel size 3–5 days, MaxPooling1D, Flatten, Dense
⑤ Hybrid CNN–LSTM — Conv1D → MaxPooling1D → Flatten → LSTM(64) → Dropout(0.2) → Dense(1, linear)

Training & Validation

80% train / 20% test ensuring test set includes crisis periods. Rolling-window cross-validation to prevent look-ahead bias. Early stopping (up to 50 epochs). Framework: Python 3.9, TensorFlow/Keras.

Evaluation Metrics

MAE — average absolute deviation (equal weight to all errors)
RMSE — penalises large errors more strongly (critical for financial risk)
MAPE — percentage error (interpretable for practitioners)

4 Results & Analysis

Table 1 — Full Model Comparison (MAE · RMSE · MAPE)

Model	MAE	RMSE	MAPE
ARIMA(1,1,1)	0.0086	0.0114	1.21%
GARCH(1,1)	0.0079	0.0103	1.09%
LSTM	0.0072	0.0094	0.97%
CNN	0.0075	0.0096	0.99%
Hybrid CNN–LSTM ★	**0.0065**	**0.0087**	**0.91%**

★ Hybrid CNN–LSTM: >20% RMSE improvement over ARIMA · Best across all three metrics · Resilient through oil crash & COVID-19

Figure 1 — RMSE Comparison Across All Models

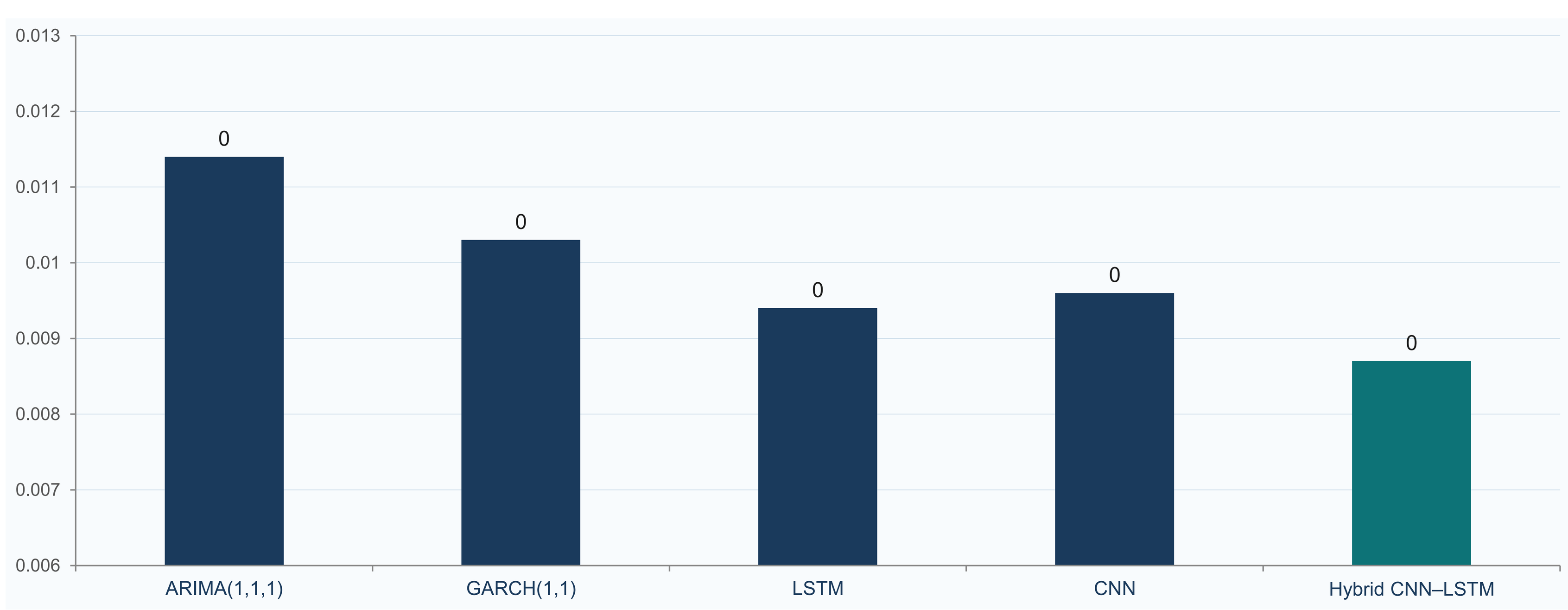

Table 2 — Robustness Checks (Hybrid CNN–LSTM)

Configuration	MAE	RMSE	MAPE
30-day window	0.0068	0.0090	0.93%
60-day (optimal)	**0.0065**	**0.0087**	**0.91%**
90-day window	0.0067	0.0089	0.92%
COVID-19 Subsample	0.0070	0.0092	0.95%

5 Conclusion

Key Findings

- Hybrid CNN–LSTM outperformed all models on MAE, RMSE, and MAPE — achieving >20% RMSE reduction vs. ARIMA
- The model proved resilient during crisis periods (oil crash 2014–15, COVID-19 2020) where econometric models failed
- CNN captures short-term volatility bursts; LSTM retains medium-to-long-range sequential dependencies — together they outperform either alone

Contributions

① First rigorous hybrid CNN–LSTM application to TASI spanning a full decade (2014–2024)
② Longest empirical horizon in Saudi stock-market deep learning forecasting
③ Multi-metric evaluation (MAE, RMSE, MAPE) with rolling-window cross-validation
④ Practical decision-support tool aligned with Saudi Vision 2030 financial innovation agenda

6 Future Work & Keywords

Future Directions

- Integrate macroeconomic variables: oil prices, interest rates, global indices
- Extend to sectoral TASI indices and individual stocks within the Saudi market
- Explore Transformer-based architectures for long-range dependency modelling
- Apply SHAP values and attention mechanisms for model interpretability
- Ensemble hybrids combining CNN–LSTM with GARCH volatility estimates

Keywords

TASI · Hybrid Deep Learning · CNN–LSTM · ARIMA · GARCH · Financial Time Series · Emerging Markets · Volatility Clustering · Saudi Vision 2030 · Rolling-Window Validation · Log-Returns